CLINICAL · NECK, THYROID & SALIVARY

Skin cancer and melanoma of the head and neck

The face, scalp and ears take the most sun, so the head and neck carry most skin cancers. Basal cell carcinoma is common and slow and almost never spreads, but it destroys locally. Cutaneous squamous carcinoma can metastasise to the parotid and neck and can track along nerves. Merkel cell carcinoma and melanoma are the dangerous minority. For all of them, biopsy first, clear margins before complex reconstruction, and think about the parotid and neck nodes.

AT A GLANCE

- Mohs surgery for high-risk and central-face BCC.
- cSCC of the scalp, temple and ear spreads to the parotid.
- Melanoma margins: 1 cm up to 1 mm thick, 2 cm over 2 mm.
- Sentinel node biopsy from T1b; checkpoint inhibitors for advanced disease.

1 CHAPTER ONE Overview

Who gets skin cancer.

1.1 The cancers

Basal cell carcinoma is the commonest human cancer and makes up about 70 to 75% of non-melanoma skin cancers; **cutaneous squamous cell carcinoma** accounts for about 20%. **Merkel cell carcinoma** is rare and aggressive. Between a fifth and a third of **melanomas** arise in the head and neck, and their incidence is rising. Rarer tumours include sebaceous carcinoma of the eyelid, adnexal carcinomas, dermatofibrosarcoma protuberans, angiosarcoma of the scalp in the elderly, and atypical fibroxanthoma.

1.2 Risk factors

Ultraviolet radiation is the dominant cause, but in two patterns: cumulative exposure drives BCC and SCC, while intense intermittent exposure and *blistering*

sunburn in childhood drive melanoma. Tanning beds add to both. Fair skin of Fitzpatrick types I and II, red or blond hair, freckling and many or atypical naevi mark the susceptible patient.

Immunosuppression transforms the risk of SCC. Organ transplant recipients have up to a hundredfold increase, and their tumours are aggressive; HIV and chronic lymphocytic leukaemia have a similar effect. Reducing immunosuppression where possible, or switching to an mTOR inhibitor such as sirolimus, helps.

Radiation, arsenic, and chronic scars and ulcers also predispose; SCC arising in a burn scar is a *Marjolin ulcer*. Among the inherited syndromes, *xeroderma pigmentosum* impairs DNA repair and produces skin cancers in childhood; *naevoid BCC (Gorlin) syndrome*, a *PTCH1* mutation in the hedgehog pathway, gives multiple BCCs with jaw keratocysts, palmar pits and a calcified falx, and radiotherapy must be avoided; the *familial atypical mole and melanoma syndrome* is linked to *CDKN2A*; and albinism removes the protection of pigment. The *Merkel cell polyomavirus* is found in most Merkel cell carcinomas.

2 CHAPTER TWO Assessment

History, examination, biopsy.

2.1 Examination

Record the lesion's size, borders, colour, ulceration and fixation, and then examine the whole skin. Palpate the draining nodes: the *parotid nodes* for the anterior scalp, temple, forehead, cheek and ear, the *postauricular and suboccipital nodes* for the posterior scalp, and the neck. Test the cranial nerves for perineural spread, looking for facial numbness and weakness (V and VII), especially with SCC of the cheek, temple and forehead.

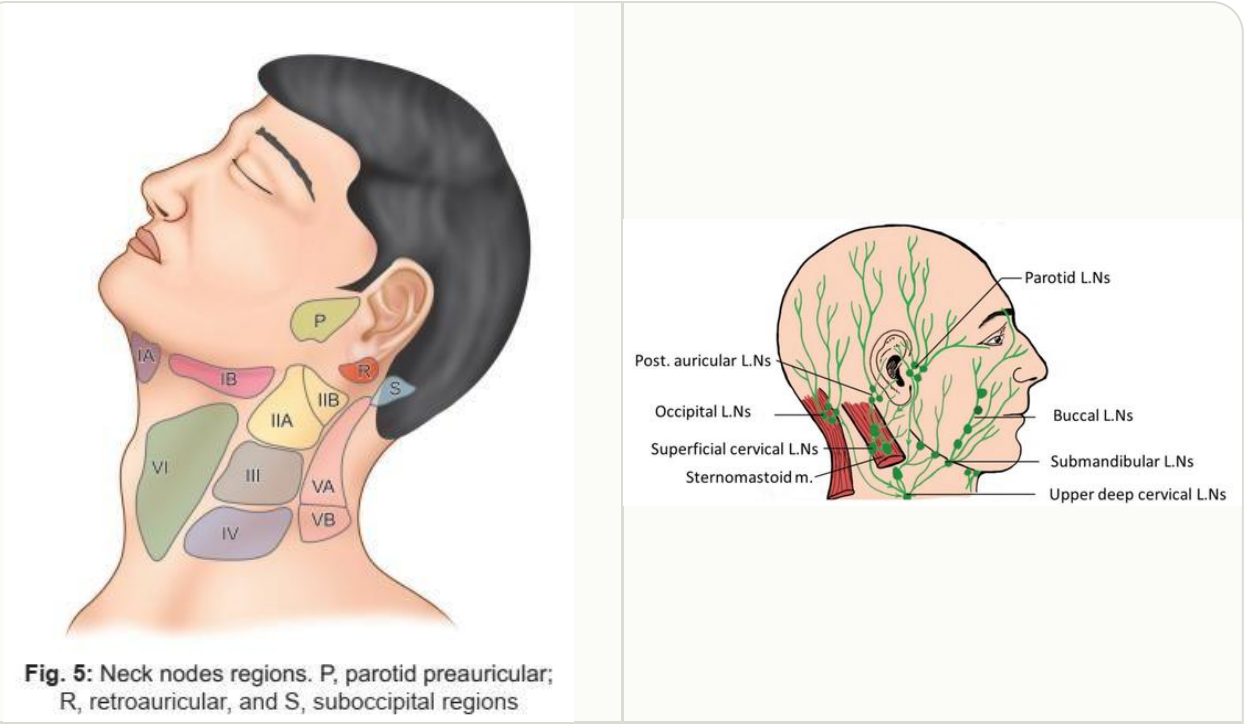


FIGURE 1 Lymph node regions of the head and neck, and the superficial nodes of the head (parotid, postauricular, occipital) that drain the scalp and face.

Anatomy atlas · ENT Steps p. 284, Doss p. 195

2.2 Biopsy

A shave biopsy suits superficial, non-pigmented lesions, and a punch or incisional biopsy through the thickest part suits larger ones. A small suspicious pigmented lesion is removed by excision biopsy with narrow margins of 1 to 3 mm. Two errors are avoided: a shave biopsy of a suspected melanoma, which loses the depth on which staging depends, and a wide excision before diagnosis, which distorts the lymphatics needed for sentinel node mapping and may be the wrong size.

2.3 Imaging

CT or MRI is used for large, recurrent or high-risk tumours, suspected bone invasion, nodes and perineural spread, with MRI following V and VII to the skull base. Nodes are assessed with ultrasound and FNA, and PET-CT stages advanced disease.

3 CHAPTER THREE Basal cell carcinoma

Locally destructive, rarely spreading.

3.1 Features

The classic BCC is a pearly, translucent papule with rolled edges and telangiectasia that ulcerates centrally, the *rodent ulcer*. The **nodular** subtype is commonest; the **superficial** type is a red scaly patch, usually on the trunk; and **pigmented** BCC can mimic melanoma. The aggressive subtypes are the *morphoeic* (sclerosing) BCC, which looks like a scar with ill-defined edges, and the infiltrative, micronodular and basosquamous types.

BCC grows slowly but invades along the embryonic fusion planes of the nasal ala, inner canthus and preauricular region, and around cartilage and bone. Metastasis is very rare.

3.2 High-risk BCC

A BCC is high risk when it lies in the *H zone* of the central face (the nose, eyelids, lips, ear, and periauricular and temple areas), when it measures 10 mm or more on the cheek, forehead, scalp or neck, or when it has ill-defined borders, an aggressive subtype or perineural invasion. Recurrent tumours, tumours in previously irradiated skin and tumours in immunosuppressed patients are also high risk.

3.3 Treatment

Low-risk BCC is excised with margins of about *4 mm*, checked histologically; small low-risk lesions off the face can be treated by curettage and cautery. **High-risk, recurrent and H-zone BCC** is best treated by *Mohs micrographic surgery*, in which horizontal frozen sections examine the whole margin, giving the lowest recurrence rates with maximum preservation of tissue; the alternative is excision with formal margin assessment before reconstruction.

Radiotherapy suits patients unfit for surgery and positive margins that cannot be re-excised, but is avoided in Gorlin syndrome and connective-tissue disease. Superficial BCC responds to topical imiquimod or 5-fluorouracil, photodynamic therapy or cryotherapy. Locally advanced or metastatic BCC is treated with *hedgehog pathway inhibitors* such as vismodegib and sonidegib, whose side effects are muscle cramps, taste loss, alopecia and weight loss, and then with cemiplimab. Because new lesions are common, patients need annual skin checks and sun protection.

CHAPTER FOUR

4 Cutaneous squamous cell carcinoma

The one that spreads to the parotid.

4.1 Spectrum

SCC develops along a spectrum. *Actinic keratoses* are premalignant rough scaly patches on sun-damaged skin, and *Bowen's disease* is SCC in situ. The *keratoacanthoma* is a fast-growing crateriform nodule with a keratin plug that may regress, but it is treated as a well-differentiated SCC and excised. Invasive SCC is a keratotic nodule or ulcer that grows over months, most often on the lip, ear, scalp, temple and cheek.

4.2 High-risk features

An SCC is high risk if it is 2 cm or more across, *more than 2 mm deep* or invading beyond the subcutaneous fat, poorly differentiated or of the desmoplastic, acantholytic or adenosquamous variant, or shows *perineural invasion*, especially of named nerves, or lymphovascular invasion. Site on the ear or lip, recurrence, origin in a scar and *immunosuppression* all add to the risk.

4.3 Treatment of the primary

Field change and SCC in situ are treated with cryotherapy, topical 5-FU or imiquimod, photodynamic therapy or curettage. Low-risk invasive SCC is excised with *4 to 6 mm margins*. High-risk SCC needs wider margins of 6 to 10 mm or Mohs

surgery with complete margin assessment, resection of underlying bone, cartilage or parotid as required, and *adjuvant radiotherapy* for perineural invasion, close or positive margins, recurrence and T3 to T4 tumours.

4.4 Nodes and the parotid

About 5% of high-risk SCCs metastasise, mostly to the *parotid* from primaries on the scalp, forehead, temple, ear and cheek, and to the neck. Nodal disease is treated by superficial or total parotidectomy with dissection of the at-risk neck levels, followed by postoperative radiotherapy, with cisplatin or a trial for extranodal extension and positive margins. Primaries on the *posterior scalp and neck* need a *posterolateral neck dissection*, which takes levels II to V together with the postauricular and suboccipital nodes. Sentinel node biopsy may be considered for selected high-risk tumours with a node-negative neck.

For advanced, unresectable or metastatic disease, the PD-1 inhibitor *cemiplimab*, and also pembrolizumab, give high response rates and have replaced platinum and EGFR inhibitors as first-line systemic treatment.

OPERATIVE GUIDES IN THEATRE 3 operations

Management of skin cancer, posterolateral neck dissection, and superficial parotidectomy.

Management of Skin Cancer of the Head and Neck
Skin cancer
Posterolateral Neck Dissection
Posterolateral neck dissection
Superficial Parotidectomy

4.5 Lip cancer

Most lip SCCs arise on the vermilion of the *lower lip* of outdoor workers; upper lip and commissure tumours are more aggressive. Lesions up to a third of the lip are removed by V or W wedge excision, and larger defects are closed with the Abbe, Estlander, Karapandzic or Bernard–Burow flaps. Actinic cheilitis is treated by vermilionectomy (a lip shave), and radiotherapy or brachytherapy is an alternative to surgery. Submental and submandibular nodes are treated when involved.

OPERATIVE GUIDES IN THEATRE 1 operation

Cancer of the lip.

Cancer of the Lip
Lip cancer

5 CHAPTER FIVE Merkel cell carcinoma

Fast and neuroendocrine.

5.1 Features and treatment

Merkel cell carcinoma affects the elderly, the sun-damaged and the immunosuppressed, and is associated with the *Merkel cell polyomavirus*. It is a rapidly growing, painless, firm, red-violet dome-shaped nodule, remembered by *AEIOU*: asymptomatic, expanding rapidly, immunosuppressed, older than 50, and on a UV-exposed site.

Histology shows small round blue cells that are *CK20 positive* in a perinuclear dot pattern and positive for neuroendocrine markers but *TTF-1 negative*, which distinguishes it from metastatic small-cell lung cancer. It spreads early to nodes, so that about a third of patients whose nodes feel normal have occult nodal disease, and it recurs locally and metastasises; five-year survival is about 30 to 65%.

Treatment is wide excision with 1 to 2 cm margins and *sentinel lymph node biopsy*, the most sensitive staging test, with neck dissection or radiotherapy for nodal disease. Adjuvant radiotherapy to the primary and nodal basin is common, and checkpoint inhibitors such as avelumab, pembrolizumab and nivolumab treat advanced disease.

6 CHAPTER SIX Melanoma

Depth is destiny.

6.1 Precursors and types

Up to a third of melanomas arise in a pre-existing naevus. Giant congenital naevi over 20 cm carry the highest risk, dysplastic naevi a lesser one, and *lentigo maligna* (Hutchinson's melanotic freckle) is melanoma in situ on sun-damaged facial skin.

Superficial spreading melanoma is the commonest, about 70%, growing first radially and then vertically. **Nodular** melanoma (10 to 15%) grows vertically from the start and is thick at diagnosis. **Lentigo maligna melanoma** appears on the face of the elderly after a long radial phase. **Acral lentiginous** melanoma of the palms, soles and nails is the commonest type in dark-skinned populations and is not sun-related. **Desmoplastic** melanoma is often amelanotic, a scar-like nodule on the head and neck that is strongly *neurotropic*, so its problem is local recurrence rather than nodes; it needs wide margins and adjuvant radiotherapy. Mucosal melanoma is considered separately below.

6.2 Recognition

The ABCDE rule describes a suspicious lesion: **A**symmetry, **B**order irregularity, **C**olour variation, **D**iameter over 6 mm, and **E**volution, meaning change in size, shape or colour, itching or bleeding. The "ugly duckling" sign, a mole unlike the patient's others, and dermoscopy add to clinical judgement.

Head and neck melanomas carry a worse prognosis than those of the trunk and limbs, the scalp being worst, then the ear, then the face and neck. The mimics are seborrhoeic keratosis, pigmented BCC, blue naevus, Spitz naevus and naevus of Ota.

6.3 Pathology and staging

Breslow thickness, the depth from the granular layer to the deepest tumour cell, is together with *ulceration* the key prognostic factor and defines the T stage. Clark's levels of anatomical invasion are now historical. The report also gives the mitotic rate, microsatellites, lymphovascular and perineural invasion and the margins. Immunohistochemistry uses S-100, SOX10, HMB-45 and Melan-A, and tumours are tested for *BRAF V600* (about half of cutaneous melanomas), NRAS, and c-KIT in mucosal and acral types.

T STAGE OF CUTANEOUS MELANOMA (AJCC 8TH EDITION)

T	BRESLOW THICKNESS	ULCERATION
Tis	In situ	—
T1a	Under 0.8 mm	Absent
T1b	Under 0.8 mm with ulceration, or 0.8 to 1.0 mm	Either
T2	Over 1.0 to 2.0 mm	a: absent; b: present
T3	Over 2.0 to 4.0 mm	a: absent; b: present
T4	Over 4.0 mm	a: absent; b: present

Stages I and II are localised disease, stage III is nodal, satellite or in-transit disease, and stage IV is distant metastasis, with the LDH forming part of the M stage. Thin melanomas need no imaging; thick or ulcerated tumours and stage III and IV disease are staged with CT or PET-CT and brain MRI, with an LDH.

6.4 Treatment of the primary

Wide local excision margins depend on thickness: 0.5 to 1 cm for melanoma in situ, 1 cm up to 1 mm thick, 1 to 2 cm for 1 to 2 mm, and 2 cm beyond 2 mm, often modified on the face to preserve function. The depth of excision depends on site: down to the periosteum on the scalp, or the outer table if it is involved; a partial or total auriculectomy on the ear, or a lateral temporal bone resection if the canal is involved; deep to the mimetic muscles on the face if needed; and to the parotid fascia, or including a superficial parotidectomy, over the parotid. Lentigo maligna is

excised in stages with margin control, by the "spaghetti" technique or Mohs with immunostains, or treated with imiquimod or radiotherapy in the frail.

DO NOT RECONSTRUCT BEFORE THE MARGINS ARE KNOWN

Complex flaps should wait until the final pathology confirms clear margins. Otherwise residual tumour is buried under a flap. Use a temporary dressing or skin graft and reconstruct afterwards (*delayed reconstruction*).

6.5 The neck

Sentinel lymph node biopsy is offered for melanomas thicker than 0.8 mm (T1b and above) with a clinically negative neck. It is the strongest predictor of survival and guides adjuvant therapy.

1. Preoperative *lymphoscintigraphy* with technetium-99m colloid, with SPECT-CT to localise nodes in the complex drainage of the head and neck, including parotid, postauricular and contralateral sites.
2. An intraoperative gamma probe, with or without blue dye.
3. Facial nerve monitoring in the parotid.

A positive sentinel node is now usually followed by *ultrasound surveillance* rather than completion dissection, because MSLT-II showed no survival benefit from completion dissection, only better regional control. Clinically or radiologically involved nodes need a therapeutic neck dissection, adding a superficial parotidectomy for primaries of the anterior scalp, temple, forehead, ear and cheek, and a posterolateral dissection with postauricular and suboccipital nodes for the posterior scalp. Adjuvant radiotherapy, often hypofractionated, improves regional control after extranodal extension or multiple or large nodes.

OPERATIVE GUIDES IN THEATRE 1 operation

Sentinel lymph node biopsy.

Sentinel Lymph Node Biopsy
Sentinel node biopsy

6.6 Systemic therapy

Checkpoint inhibitors, anti-PD-1 agents (nivolumab, pembrolizumab) alone or with the anti-CTLA-4 agent ipilimumab, are first-line treatment for unresectable and metastatic melanoma and give durable responses in about half of patients, at the cost of immune side effects: colitis, hepatitis, pneumonitis, thyroiditis and hypophysitis. **BRAF with MEK inhibitors**, dabrafenib with trametinib or encorafenib with binimetinib, give rapid responses in BRAF-mutant disease. After resection of stage III and high-risk stage II melanoma, adjuvant anti-PD-1 therapy, or BRAF/MEK inhibitors for BRAF-mutant tumours, reduces relapse. Intralesional oncolytic virus (T-VEC) treats in-transit disease, while dacarbazine and interferon are now historical.

> **THE NECK AFTER A POSITIVE SENTINEL NODE, AND ADJUVANT IMMUNOTHERAPY**
>
> Older practice was completion neck dissection for every positive sentinel node, with interferon as the only adjuvant. Since MSLT-II and DeCOG-SLT, most patients are followed with nodal ultrasound instead. Adjuvant anti-PD-1 or BRAF/MEK therapy is now standard for resected stage III disease.

6.7 Mucosal melanoma

Mucosal melanoma makes up only about 1% of melanomas, but the head and neck is its commonest site: the nasal septum and turbinates, the sinuses, the palate and gingiva, and the nasopharynx. It is often amelanotic and presents late with unilateral obstruction, epistaxis or a pigmented palatal lesion. Breslow depth does not apply, and every mucosal melanoma is at least T3.

Treatment is complete endoscopic or open resection with postoperative radiotherapy for local control, and therapeutic rather than elective neck dissection. The prognosis is poor, with five-year survival of 20 to 30% because of distant metastasis, and immunotherapy and KIT inhibitors have a role.

6.8 Melanoma of unknown primary

Melanoma may present in the parotid or neck nodes with no primary, which has perhaps regressed. Examine the skin, scalp, mucosa and eyes, then treat it as stage

III disease with dissection, adjuvant radiotherapy and systemic therapy. The prognosis is similar to, or better than, that of stage III disease with a known primary.

7 CHAPTER SEVEN Reconstruction

After excision.

7.1 The reconstructive ladder

Defects are closed by the simplest reliable method on the *reconstructive ladder*:

1. Healing by secondary intention, which works well in concave sites such as the conchal bowl and medial canthus.
2. Primary closure along relaxed skin tension lines.
3. Skin grafts: full-thickness grafts from postauricular or supraclavicular skin for the nose and eyelids, and split-thickness grafts for the scalp over periosteum.
4. Local flaps: advancement, rotation, transposition (rhomboid, bilobed), and interpolated flaps (paramedian forehead, nasolabial).
5. Regional flaps: cervicofacial, deltopectoral, pectoralis major, trapezius.
6. Free tissue transfer.

Scalp defects can also be closed by tissue expansion.

OPERATIVE GUIDES IN THEATRE 5 operations

Skin grafting, cervicofacial and paramedian forehead flaps, scalp reconstruction, and free flaps.

Skin Grafting
Skin grafting
Cervicofacial Transposition Flaps
Paramedian Forehead Flaps
Forehead flap
Scalp Reconstruction: Role of Tissue Expansion and Flap Reconstruction
Microvascular Reconstruction of the Head and Neck

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **BCC:** features, high-risk features, Mohs surgery and its rationale, hedgehog inhibitors.
- **cSCC:** high-risk features, parotid metastasis, perineural spread, cemiplimab.
- **Merkel cell carcinoma:** CK20 and TTF-1; sentinel node biopsy.
- **Melanoma:** ABCDE, Breslow and ulceration, margins by thickness, sentinel node indications and lymphoscintigraphy, MSLT-II, checkpoint and BRAF/MEK inhibitors.
- **Drainage:** of the anterior versus posterior scalp, and when to add parotidectomy.
- **Mucosal melanoma.**

8.2 Common pitfalls

COMMON PITFALLS

- Shave biopsy of a pigmented lesion.
- Wide excision before diagnosis, which ruins sentinel node mapping.
- Flap reconstruction before margins are clear.
- Missing facial numbness or weakness as perineural spread from a "treated" facial SCC.
- Radiotherapy in Gorlin syndrome.

Written for this app from K. J. Lee ch. 42, 43. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

SALIVARY GLANDS · TUMOURS

Malignancies of the Parotid Gland

IMAGING FINDINGS

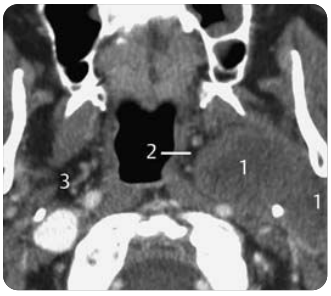
- Ill-defined or infiltrative margins
- Low or intermediate T2, low ADC
- Facial nerve: enhancement through the stylomastoid foramen

TOP DIFFERENTIALS

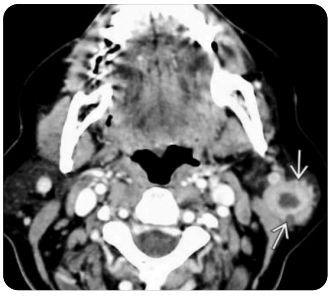
- Pleomorphic adenoma
- Warthin tumour
- Metastatic node from skin cancer; lymphoma

CHECKLIST

- Perineural spread to the mastoid segment and beyond
- Nodes in the parotid and level II



Slight swelling of the left parotid region and deep pain.
Dunnebier, Imaging for Otolaryngologists, Fig. 9.30 a



Axial CECT reveals a well-defined mass in the superficial lobe of the parotid gland. It has a thick rind of peripheral enhancement and is centrally cystic …

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 98.